Dietary fiber and breast cancer risk: a systematic review and meta-analysis of prospective studies.

BACKGROUND: Evidence from case-control studies suggest that dietary fiber may be inversely related to breast cancer risk, but it is unclear if this is supported by prospective data. We conducted a systematic review and meta-analysis of the evidence from prospective studies. METHODS: PubMed was searched for prospective studies of fiber intake and breast cancer risk until 31st August 2011. Random effects models were used to estimate summary relative risks (RRs). RESULTS: Sixteen prospective studies were included. The summary RR for the highest versus the lowest intake was 0.93 [95% confidence interval (CI) 0.89-0.98, I(2) = 0%] for dietary fiber, 0.95 (95% CI 0.86-1.06, I(2) = 4%) for fruit fiber, 0.99 (95% CI 0.92-1.07, I(2) = 1%) for vegetable fiber, 0.96 (95% CI 0.90-1.02, I(2) = 5%) for cereal fiber, 0.91 (95% CI 0.84-0.99, I(2) = 7%) for soluble fiber and 0.95 (95% CI 0.89-1.02, I(2) = 0%) for insoluble fiber. The summary RR per 10 g/day of dietary fiber was 0.95 (95% CI 0.91-0.98, I(2) = 0%, P(heterogeneity) = 0.82). In stratified analyses, the inverse association was only observed among studies with a large range (≥13 g/day) or high level of intake (≥25 g/day). CONCLUSION: In this meta-analysis of prospective studies, there was an inverse association between dietary fiber intake and breast cancer risk.